



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS FALL 2022

Faculty	Computing and Applied Sciences	Department	Computer Science & Information Technology	Batch	2021F
Semester	Fall 2022	Degree Program	BS (CS)	Course Instructor	Adeel Saeed, M. Fahim
Course Code	CS-212T	Course Title	Data Structures and Algorithms	Credit Hours	3
Date of Examinations	4 th Feb 2023	Timings	9:30am – 12:30pm	Shift	1 st

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.	2021F-BCS-009	Name	Ahsan Fayyaz	Section	A	'A' Copy Serial No.	128412
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Q.No.1

(5+5)

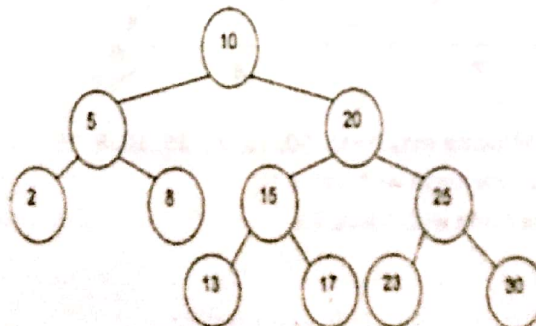
(a) Does the pre-order traversal of two different trees be the same even though they are different? If it is, show with an example.

(b) Using inorder, preorder, and postorder tree traversal, visit only leaves of a tree. What can you observe? How can you explain this phenomenon?

Q.No.2

(5+5)

(a) Insert 18 into the following AVL tree. What type of imbalance does it cause? Show the result after balancing.





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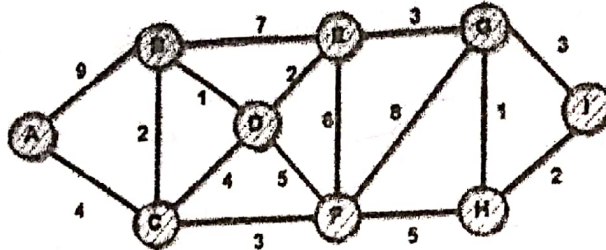
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- ✓ (b) A file contains the following characters with the frequencies as shown. If Huffman Coding is used for data compression, determine Huffman Code for each character and length of Huffman encoded message (in bits).

Characters	a	e	i	o	u	s	t
Frequencies	10	15	12	3	4	13	1

Q.No.3 (5+5)

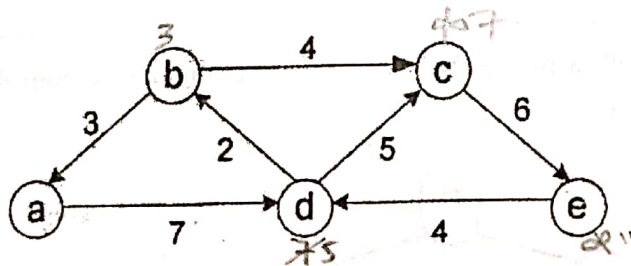
- (a) Compute the Minimum Spanning Tree (MST) of the following graph using Prim's algorithm. Show all steps.



- ✓ (b) For the graph used in 3(a), compute the Minimum Spanning Tree (MST) using Kruskal's algorithm. Show all steps.

Q.No.4 (5+5)

- ✓ (a) Find the shortest distance from source vertex 'a' to remaining vertices in the following graph using Dijkstra's Algorithm?



- ✓ (b) Consider the following max heap 50, 30, 20, 15, 10, 8, 16
- Insert a new node with value 60.
 - Delete a node with value 50.



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Q.No.5

(5+5)

- ✓ (a) Given a Binary Search Tree, write an efficient function to search a given key in it. The algorithm should return the parent node of the key and print if the key is the left or right node of the parent node. If the key is not present in the BST, the algorithm should be able to determine that.
- ✓ (b) Consider an array of elements: 8, 3, 2, 7, 9, 1, 4
- Draw a max heap to represent the array.
 - Implement a heap sort in the given array to display the initial list and sorted list.